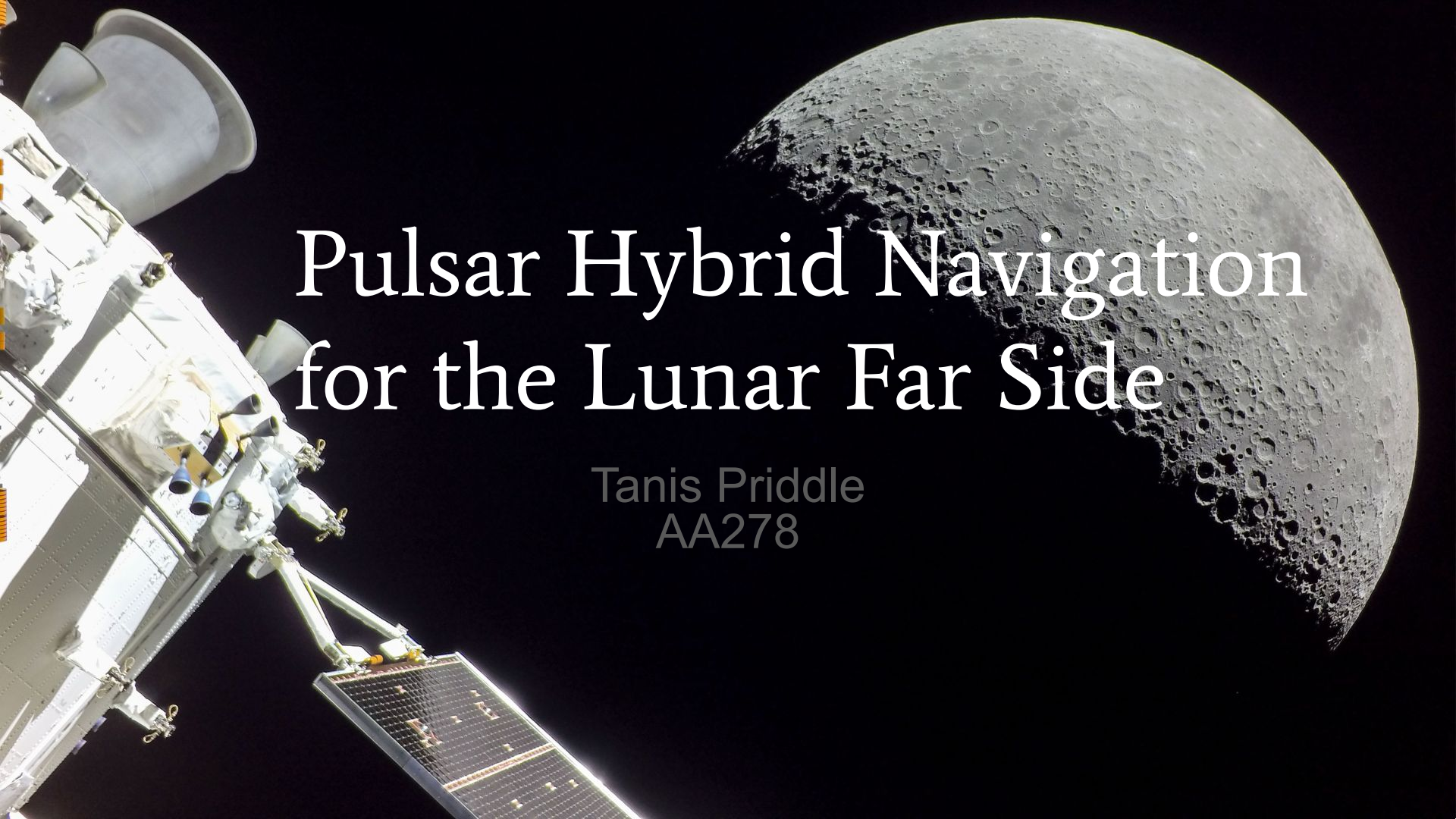


A photograph of a satellite in space, likely the Lunar Reconnaissance Orbiter, with its solar panels extended. The Moon is visible in the background, showing its heavily cratered surface. The satellite is white with various instruments and antennas. The background is the black of space.

Pulsar Hybrid Navigation for the Lunar Far Side

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Design and evaluate a hybrid EKF that continuously fuses x-ray pulsar time-of-arrival measurements with LunaNet pseudorange fixes when available to maintain bounded position and timing accuracy for a lunar orbiter through far-side blackout periods where no Earth GNSS sidelobe signals are available and near-side navigation under poor sidelobe geometry.

- The lunar far side is permanently shielded from Earth -> no GNSS sidelobe signals reach it, leaving far-side orbiters with no absolute PNT source during blackouts that can last hours per orbit
- Far-side missions are a reality (Artemis II woohoo!) and scientifically critical, but the current ODS achieves only 1.5 km accuracy even on the near side, which is unacceptable for the LunaNet 13.43 m^[1] requirement
- X-ray pulsars are a natural absolute positioning source available anywhere in deep space, are permanently visible regardless of orbital phase, and require no additional lunar infrastructure!

Proposed Approach

Simulator Setup

Propagate a far side lunar orbiter (30 hr ELFO or 2 hr LLO) using the ODTS dynamics model from lecture, including high-order lunar gravity, Earth/Sun third-body perturbation, and SRP, to generate a ground-truth trajectory

$$\mathbf{r}_k, \mathbf{v}_k, c\delta t, c\delta\dot{t}, c\delta\ddot{t}, \gamma$$

Visibility Modeling

Find LunaNet satellite and Earth GNSS visibility windows in reference to orbital phase $\rightarrow$ label far side blackout periods

Pulsar Measurement Model

Use a catalog of 3-5 bright millisecond pulsars. Model each XNAV measurement as a scalar observable (the dot product of the spacecraft position error with the unit vector toward each pulsar) with Gaussian noise parameterized by timing uncertainty.

$$\begin{aligned}\rho^{(1)} &= \sqrt{(x^{(1)} - x)^2 + (y^{(1)} - y)^2 + (z^{(1)} - z)^2} + b_u - B^{(1)} \\ \rho^{(2)} &= \sqrt{(x^{(2)} - x)^2 + (y^{(2)} - y)^2 + (z^{(2)} - z)^2} + b_u - B^{(2)} \\ \rho^{(3)} &= \sqrt{(x^{(3)} - x)^2 + (y^{(3)} - y)^2 + (z^{(3)} - z)^2} + b_u - B^{(3)} \\ \rho^{(4)} &= \sqrt{(x^{(4)} - x)^2 + (y^{(4)} - y)^2 + (z^{(4)} - z)^2} + b_u - B^{(4)}\end{aligned}$$

$$c(\delta t_b - \delta t_{\text{obs}}) - \delta \hat{\mathbf{n}} \cdot \tilde{\mathbf{r}} = \tilde{\mathbf{n}} \cdot \delta \mathbf{r}$$
$$\Delta \rho = \begin{bmatrix} \hat{\mathbf{n}}_1 & 1 \\ \hat{\mathbf{n}}_2 & 1 \\ \vdots & \vdots \\ \hat{\mathbf{n}}_j & 1 \end{bmatrix} \begin{bmatrix} \mathbf{r} \\ c\delta t_c \end{bmatrix} = \mathbf{H}' \mathbf{r}'$$

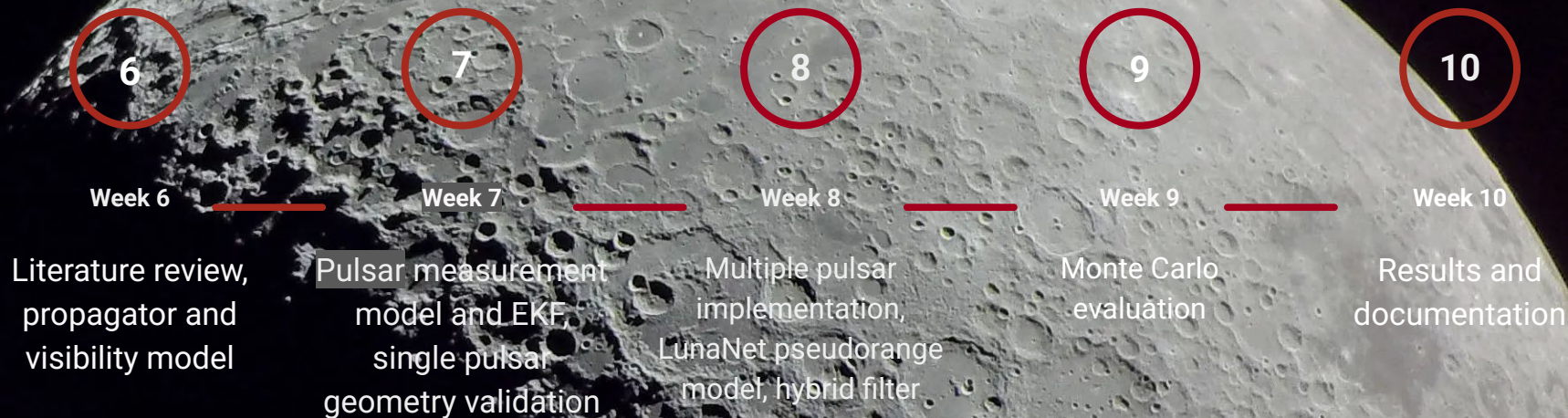
Filter Design

Implement an EKF with 10-state vector; using XNAV measurements during blackout phases and supplementing LunaNet pseudoranges when visible

Evaluation

Run Monte Carlo simulations over blackout duration, number of pulsars, and timing noise level. Compare position and timing SISE with the LunaNet requirement for three configurations: 1. sidelobe-only, 2. XNAV-only, 3. hybrid

Timeline



Literature Review

- Standalone GPS L1 C/A Receiver for Lunar Missions (Capuano et al., 2016): Developed an adaptive EKF orbital filter fusing GPS L1 C/A pseudoranges from sidelobe signals for lunar orbit navigation and demonstrated 10-100 m accuracy in simulation
- GNSS-based Lunar Orbit and Clock Estimation With Stochastic Cloning UD Filter (Iiyama & Gao, 2026): Proposed the SC-UDU filter with TDCP and full dynamics model, demonstrating meter-level accuracy in Monte Carlo simulation
- Spacecraft Navigation Using X-Ray Pulsars (Sheikh & Pines, 2006): Established the theoretical basis for X-ray pulsar navigation, including time-of-arrival measurement model, scalar range projection along pulsar line-of-sight, and navigation performance analysis for spacecraft
- Pulsar Selection Criteria and Performance Evaluation of Autonomous X-ray Pulsar Navigation Systems (Chen et al., 2025): Analysis of pulsar selection criteria and EKF-based XNAV performance using NICER observational data; derives TOA noise estimates and evaluates navigation accuracy for LEO and interplanetary scenarios
- NICER/SEXTANT Demonstration (NASA 2018): First on-orbit autonomous X-ray pulsar navigation demonstration aboard the ISS; achieved accuracy to within 3 miles using millisecond pulsars